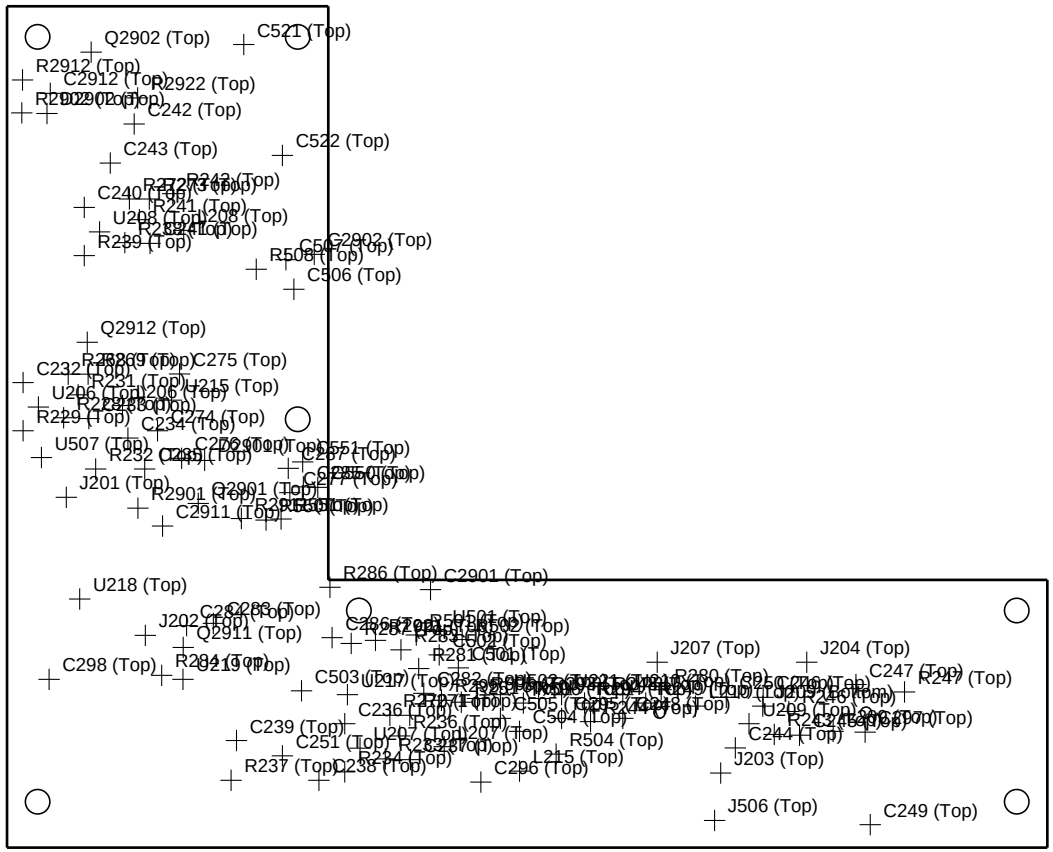

Board extent 136.000 mm (X) x 110.000 mm (Y)

Top view, Edge.Cuts and holes of PWR_B.kicad_pcb; crosses = CPL positions (KiCad footprint origins). Not to a fixed scale.
NOT RELEASED: model/CI only; bench gates OPEN

PWR_B fab notes

NOT RELEASED: model/CI only; bench gates OPEN.
Board status: MODEL_NOT_FOR_FABRICATION. Every value below is a CI/model output of pcb/fab_export.py and pcb/board_gen.py; no hardware or bench gate is closed by it.

Board

Outline (Edge.Cuts of PWR_B.kicad_pcb): 136.000 x 110.000 mm, area 7910.0 mm2; lower-left corner at X 100.000, Y -210.000 mm in the Gerber/CPL frame (KiCad absolute origin, Y up).
Holes: NPTH 1.45 mm, NPTH 3.20 mm (9 mounting holes; NPTH unplated and without net).

Fabrication

Copper layers: 6; finished thickness 1.6 mm (JLC tolerance +/-10 % for boards >= 1.0 mm).
Stack-up JLC06161H-2116 (https://jlcpcb.com/impedance, read 2026-11-12; pcb/generated/stackup-6l.json (pcb/stackup.py)):
L1 F.Cu: copper 0.07 mm
prepreg 2116x1 + 2313x1: 0.2234 mm, Er not given on the source page
L2 In1.Cu: copper 0.0152 mm
core: 0.3 mm, Er 4.6
L3 In2.Cu: copper 0.0152 mm
prepreg 7628x1 + 7628x1: 0.4168 mm, Er 4.4
L4 In3.Cu: copper 0.0152 mm
core: 0.3 mm, Er 4.6
L5 In4.Cu: copper 0.0152 mm
prepreg 2313x1 + 2116x1: 0.2234 mm, Er not given on the source page
L6 B.Cu: copper 0.07 mm
Layer thicknesses sum to 1.5944 mm (computed). Copper: outer 0.035 mm = 1 oz, inner 0.0152 mm = 0.5 oz finished (the inner default on jlcpcb.com/capabilities/pcb-capabilities: "Finished copper weight of inner layer is 0.5oz by default"), so the inner widths of netclasses.json (0.5 oz) hold unchanged. Er of 2313 is not on the page (None): the 6-layer board carries no RF50 net, so no impedance is modelled or ordered for it. Copper option outer_2oz: finished outer copper 0.07 mm (2 oz), inner 0.0152 mm (0.5 oz), read 2026-11-13; the dielectric table of JLC06161H-2116 with 0.070 mm outer foil sums to 1.6644 mm (computed), inside the 1.6 mm +/-10 % order tolerance; the fab sets the finished build (not on the pages read): TBD at order; price TBD: the copper options carry no price on the pages read; the JLC quote calculator was not read.
6 layers, stack-up JLC06161H-2116 (https://jlcpcb.com/impedance, read 2026-11-12): JLCPCB Via-in-Pad Process: vias filled with epoxy resin or copper paste and plated over; "This process is the default for 6-layer and above multilayer boards"; compatible with via diameters 0.15 to 0.55 mm (https://jlcpcb.com/capabilities/pcb-capabilities, read 2026-11-12); surface finish ENIG: jlcpcb.com/capabilities/pcb-capabilities (Surface Finish, read

2026-11-13) offers HASL (leaded / lead-free), ENIG and OSP, and "FR-4/HDI boards with 6 or more layers" do not support HASL; order without impedance control (no RF50 net on this board). Order finished outer copper 2 oz and inner 0.5 oz (outer_2oz), stack-up JLC06161H-2116; finished thickness TBD at order: the layer table with this copper sums to 1.664 mm (MODEL) against the nominal 1.6 mm +/-10 %.
Impedance: no RF50 net on this board; order without impedance control.
Surface finish, solder mask and legend colours: order options, not fixed by this fab set.
KiCad 10.0.7 DRC on the committed board: 0 errors, 0 unconnected, 398 reviewed warnings, ignored checks [] (report.json; CI/model result).

Files (fab/)

gerber/: Gerber X2 with netlist attributes, Protel extensions, one file per layer, Gerber job file (.gbrjob). drill/: Excellon PTH and NPTH files (mm, decimal, absolute origin, alternate oval mode) and Gerber X2 drill maps. PWR_B.ipc356: IPC-D-356 netlist of every net with a pad. assembly/: JLC BOM and CPL, the raw KiCad position file, the rotation offset table and the instance map.

Assembly

Boards to build: 1 = instances (PWRB 1) + spares 0; every board is the same design and is assembled the same.
Sides with parts: Bottom (SMD 1), Top (SMD 122, THT 6).
CPL: Mid X / Mid Y are the KiCad footprint origins (JLC defines them as the component centroid) and Rotation the raw KiCad rotation (degrees, counter-clockwise as the JLC CPL column). JLC rotation and centroid offsets per footprint are TBD (assembly/PWR_B-cpl-rotation-offsets.csv): check every part in the JLC viewer at order time; nothing here verifies them.
BOM TBD: C274,C275,C276,C277,C283,C285,C287,C2911,C2912 (C1608X7R1H104K080AA): TBD: build/10-jlc-bom.csv row has no LCSC code

Not proved here (OPEN)

- that the schematic is right
- current capacity or temperature rise beyond the chosen class widths
- impedance, RF loss or isolation, EMC or signal integrity
- creepage beyond the encoded clearances
- that the enclosure fits (only outline and hole checks; heights are not checked here)
- that the fab's DFM accepts the board, or that CPL rotations are right
- that the land patterns match the purchased MPNs
- any function: hardware and bench gates stay OPEN

Sources (read 2026-11-07): https://jlcpcb.com/help/article/bill-of-materials-for-pcb-assembly; https://jlcpcb.com/help/article/pick-place-file-for-pcb-assembly; https://jlcpcb.com/help/article/how-to-generate-the-bom-and-centroid-file-from-kicad; https://jlcpcb.com/help/article/how-to-generate-gerber-and-drill-files-in-kicad-8; https://jlcpcb.com/capabilities/pcb-capabilities